

Swiss Institute of Banking and Finance



University of St.Gallen

11. Introduction to Empirical Research in (Corporate) Finance

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3,130/4,130 Corporate Finance

Course outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
4. Corporate Valuation
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
- 11. Empirical Research in Corporate Finance**

Agenda

1. Empirical corporate finance / governance research: The empirical setup
 - How to measure performance?
 - How to measure corporate governance?
 - Data availability
 - The empirical setup
2. Empirical evidence:
 - Do forced CEO turnovers hurt involved directors' reputation?

How to measure performance?

Tobin's Q:

- The most often used measure of firm valuation / performance is Tobin's Q (Tobin, 1969).
- Tobin's Q is defined as the ratio of the market value of a company to its replacement (or recorded asset) value.
- In an empirical (finance) setup it is common to use the following simple approximation:

$$Q = \frac{(\text{Market Value of Equity} + \text{Book Value of Debt})}{\text{Total Assets}}$$

- If the market valuation reflected solely the recorded assets of a company, Tobin's Q would be 1.0.

How to measure performance? (II)

- If Q is greater than 1.0, the market value is greater than the value of the company's recorded assets, suggesting that the market value reflects some unrecorded assets of the company.
- How is corporate governance reflected in Tobin's Q ?:
 - In better governed firms, less assets, profits, etc. are expected to be diverted by self-serving managers as compared to poorly governed companies.
 - In an efficient market (and due to transparency requirements), the market should realize this and hence be willing to pay more for well-governed companies leading to higher share prices.
 - Hence, the numerator in the Q formula increases while the denominator remains unchanged (see governance class on why the denominator may change as well) leading to higher Q -values.

How to measure performance? (III)

An alternative are stock returns, but they are associated with several problems:

- In an efficient market, we would expect that only changes in corporate governance affect stock returns (at announcement): Changes vs. levels!
- However, if governance matters but is not incorporated immediately into stock prices, then realized returns on the stock would differ systematically from equivalent securities.
- Which is the relevant time window? Example: Aebi, Sabato, and Schmid (2012) test whether bank performance was different during the financial crisis depending on the pre-crisis (risk) governance.
- Joint test of the risk model and “governance story”: What is the correct factor model (e.g., Fama and French, 1993; Carhart, 1997)?

How to measure performance? (IV)

Accounting-based measures (examples):

- Return on assets (e.g., EBIT / total assets)
- Return on equity (e.g., net income / book value of equity)
- Sales growth, asset growth, etc.

⇒ Major problem: These measures are all *backward-looking*.

Other measures of company performance:

- Value destroying (or increasing) acquisitions and value-increasing (or destroying) divestitures
- More valuable (e.g., R&D) investments, more patents, etc.
- Positive news and press coverages (measured using text analysis)

How to measure corporate governance?

Earlier studies investigate the effect of individual and “simple” corporate governance attributes:

- Morck, Shleifer, and Vishny (1988) show that managerial ownership is related to firm value in a non-linear way: positive for ownership between 0% and 5%, negative for an ownership range between 5% and 25%, and positive beyond 25%.
- Hermalin and Weisbach (1991) show that, controlling for manager ownership, the percentage of outside directors on the board is not significantly related to firm value.
- Yermack (1996) shows that board size is negatively related to firm value (and controls for manager ownership and the percentage of outside directors on the board).

How to measure corporate governance? (II)

More recent studies attempt to use more comprehensive measures of corporate governance:

- Gompers, Ishii, and Metrick (2003) use an index that comprises 24 different corporate governance attributes that are related to anti-takeover provisions (G-Index).
- Bebchuk, Cohen, and Ferrell (2009) propose a reduced version of the G-Index which consists of six out of the 24 attributes with the empirically strongest relation to firm value and returns (E-Index).
- Ammann, Oesch, and Schmid (2011) use 64 corporate governance attributes from the Governance Metrics International (GMI) database. The sample includes 6,600 firm-years on almost 2,000 firms from 2003 to 2007.

How to measure corporate governance? (III)

Board Accountability

1. Board members are subject to annual election by all shareholders	22.2%
2. Non-executive board members have a formal session without executives once a year	55.3%
3. Board performance is periodically evaluated	86.2%
4. Company discloses a code of ethics for senior executives	45.3%
5. Company discloses its corporate governance policies or guidelines	64.2%
6. Board or a committee is responsible for CEO succession planning	82.8%
7. Company has not failed to adopt the recommendations of a shareholder proposal	99.8%
8. All executive board members own shares after excluding options held	73.1%
9. All non-executive board members own shares after excluding options held	37.5%
10. Company has a separated chairman and CEO	86.0%
11. All members attended at least 75% of the board meetings	80.5%
12. Company has a designated "lead" or senior non-executive board member	28.5%
13. There have been no related-party transactions in the past three years	50.0%
14. The governance/nomination committee is composed of independent board members	20.9%
15. No former CEO of the company serves on the board	72.2%
16. Nr. of shares held by officers and directors has not decreased by 10% or more	84.2%
17. Nr. of shares held by officers and directors has increased by 10% or more	26.0%
18. Governance/nomination committee has a written charter or terms of reference	50.6%
19. Board size is greater than five but less than 16	80.5%
20. Board is controlled by more than 50% of independent outside directors	40.7%

How to measure corporate governance? (IV)

Financial Disclosure and Internal Control

21. Company has not had a material earnings restatement in the past three years	98.3%
22. Audit committee has a written charter or terms of reference	63.9%
23. Company has not received a qualified audit opinion within the last two fiscal years	99.4%
24. Company is not currently under investigation for accounting irregularities	98.8%
25. Audit committee is wholly composed of independent board members	43.8%
26. Someone other than senior management with sole authority to hire outside auditor	85.1%
27. Audit committee with sole authority to approve non-audit services from outside auditor	41.9%
28. Company did not pay its auditor less for audit related services than for other services	84.4%

Shareholder Rights

29. Vote results for the last shareholder meeting are disclosed within 14 calendar days	78.5%
30. All common or ordinary equity shares have one-share, one-vote, with no restrictions	68.0%
31. The company provides confidential voting with no or with reasonable exceptions	50.5%
32. Shareholders have a right to convene an EGM with 10% or less of the shares requesting one	90.8%
33. Shareowners have a right to act in concert through written communication	17.6%
34. Voting rights are not capped at a certain percentage	95.4%

Remuneration

35. Non-executive board members paid in cash and some form of stock-linked compensation	18.1%
36. Company discloses performance targets for the next fiscal year	31.1%
37. Non-executive board members are paid entirely in some form of stock-linked compensation	0.5%

How to measure corporate governance? (V)

38. CEO without an employment agreement that provides for guaranteed bonus payments	97.9%
39. Goals used to determine incentive awards are aligned with the company's financial goals	44.7%
40. CEO/Managing Director does not sit on the remuneration committee	94.9%
41. Remuneration committee is wholly composed of independent board members	35.9%
42. No repricing of outstanding executive stock options and no option exchange program	98.9%
43. Expensing of employee stock option grants	35.5%
44. Remuneration committee has a written charter or terms of reference	56.4%
45. Potential Dilution from Stock Options Outstanding is below 20%	99.8%
46. Potential Dilution from Stock Options Outstanding + not yet granted is below 20%	99.5%

Market for Control

47. Company has not adopted a shareholder rights plan ("poison pill")	96.3%
48. Company does not have a staggered ("classified") board	51.4%
49. Company cannot issue blank check preferred stock in the event of a hostile tender offer	89.5%
50. Company's shareholder rights plan ("poison pill") has been ratified by a shareholder vote	3.4%
51. Fair price provision in place or price protection under applicable law	69.9%
52. Shareholder rights plan includes a TIDE provision or a three-year sunset provision	1.1%
53. Company does not require a supermajority vote to approve a merger	42.7%
54. No single shareholder or shareholder group with majority of voting power	81.3%
55. Company allows cumulative voting in the election of directors	3.8%

How to measure corporate governance? (VI)

Corporate Behavior

56. The company have a policy addressing workplace safety	86.0%
57. Company does not have pending criminal litigation against it	96.4%
58. No allegation that the company used sweat shops within the last three years	99.8%
59. Company discloses its environmental performance	52.2%
60. Company discloses its workplace safety record	34.7%
61. No regulatory investigation for a material issue other than for accounting irregularities	91.9%
62. Company discloses its policy regarding corporate level political donations	27.5%
63. Company has not been charged with workplace safety violations within the last two years	98.7%
64. It has not been alleged by a responsible party that the company used child labor	99.7%

Source: Ammann, Oesch, and Schmid (2011)

Data availability

Reliable / usable data is available for the following countries / firms:

- Managerial compensation and ownership data is available for US firms (S&P 1500) from S&P's ExecuComp database for the years 1992 onwards.
- Detailed data on board characteristics is available from the ISS (formerly RiskMetrics) Directors database for S&P 1500 firms since 1996.
- Data to calculate the Gompers, Ishii, and Metrick (2003) index (plus some additional governance variables) are available from the ISS Governance database for the years 1990 to today (1990, 1993, 1995, 1998, 2000, 2002, 2004, 2006, 2007-).

Data availability (II)

- Detailed data on virtually all directors in listed US firms and many directors worldwide is available on BoardEx (as of 2000).
 - Governance Metrics International provides about 400 corporate governance attributes on thousands of international companies.
 - Shareholder voting data (on say-on-pay, director elections, etc.) is available in ISS Voting Analytics.
 - Data on block shareholdings (holdings exceeding 5%) are available from Thomson Financial's CDA/Spectrum database.
 - Thomson's Insider Trading database includes information on stock and option trades of top executives and directors since 1986.
- ⇒ Most of these datasets are available on WRDS (Wharton Research Data Services) and thus widely available to researchers.

Data availability (III)

Unfortunately, there is still only very limited data on non-US countries:

- For a long time, most papers used hand-collected data from annual reports and/or survey data.
- The emergence of ISS and GMI (and a few others) made large-scale international studies possible. But these databases do not include detailed information on managerial ownership and remuneration and are very expensive.
- Recently, Thomson Financial started to collect and report a number of board characteristics (board size, % of independent directors on the board, etc.) on international firms as well.
- BoardEx also added more and more countries and firms.

Data availability (IV)

- Most data is available in yearly intervals only.
 - The exact date of changes in governance is in general not reported in databases and the announcement often coincides with other major events that are reported on the same day (e.g., at the Annual General Shareholder Meeting).
 - Hence, to be able to investigate the announcement effects of important changes in governance, a detailed news search for example in LexisNexis or in Dow Jones Factiva is necessary.
- ⇒ Basically all older empirical studies use either cross-sectional or panel datasets including yearly observations.
- ⇒ More recent research looks at changes in corporate governance (and their announcement effects), changes in regulation, etc.

The empirical setup

- McConnell and Servaes (1990) estimate the following regression model:

$$Q = \alpha + \beta_1 \cdot INOWN + \beta_2 \cdot INOWN^2 + \beta_3 \cdot INSTO + \beta_4 \cdot DEBT / ASSETS + \beta_5 \cdot R\&D / ASSETS + \beta_6 \cdot ADV / ASSETS + \beta_7 \cdot ASSETS + \varepsilon$$

where:

- INOWN = Stock ownership by insiders (officers and directors)
- INSTO = Institutional ownership
- DEBT/ASSETS = Debt divided by total assets (=leverage)
- R&D = Research and development expenditures
- ADV = Advertising expenditures

The empirical setup: Refresher

- To assess the statistical significance of results, we usually use t-statistics, defined as the coefficient estimate divided by the standard deviation of the estimate.
- Values larger than 1.645 indicate statistical significance at the 10% level, larger than 1.96 at the 5% level, and 2.58 at the 1% level.
- Alternatively, we report p-values: The p -value is the probability of obtaining the observed sample results, or "more extreme" results, when the null hypothesis is actually true. p -values directly represent the significance level: ≤ 0.10 , ≤ 0.05 , and ≤ 0.01 .
- *, **, *** usually indicate significance at the 10%, 5%, 1% level.

Variable

(1)

Entrenchment Index E

-0.044***
0.004

t-statistic: $-0.044 / 0.004 = 11$

p-value: ≤ 0.01

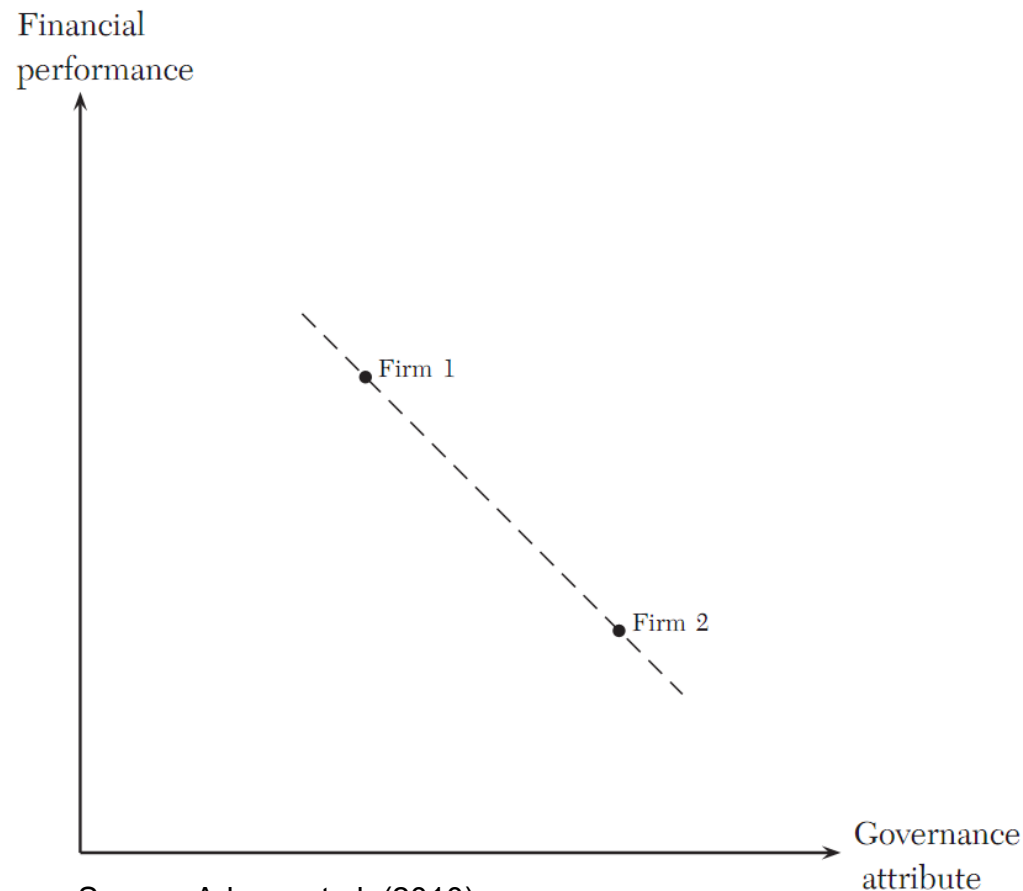


The empirical setup: Refresher (II)

- Multivariate (or multiple) regressions allow us to estimate the effect of one variable (e.g., board size) on firm value while holding all the other variables (e.g., board independence, CEO ownership, etc.) constant.
- Adding time (usually year) and industry dummy variables, additionally controls for general time trends and industry effects.

The empirical setup: Heterogeneity

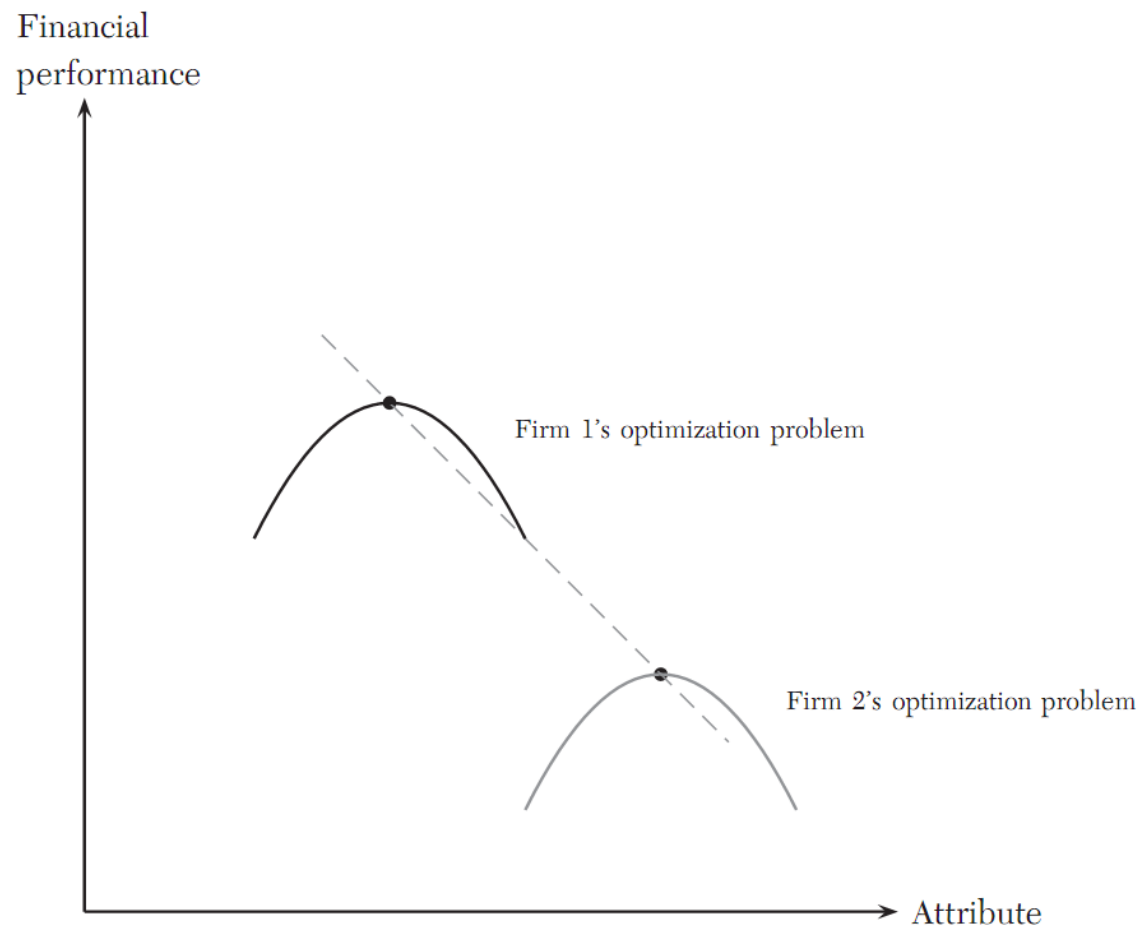
- The relation between a specific firm-attribute (e.g., board size) and firm performance / value:



Source: Adams et al. (2010)

The empirical setup: Heterogeneity (II)

- The optimization problem of the two firms:



Source: Adams et al. (2010)

The empirical setup: Heterogeneity (III)

- For both firms there is a concave relation between the governance attribute and firm value.
- In the graph, both firms are at their maximum!
- Firm 2 would prefer to be on Firm 1's curve, but it isn't.
- But: by decreasing board size, Firm 2 would do worse than it does with the current board size.
- A simple linear regression as in the first graph is misleading.

Example:

- Yermack (1996) shows that larger boards are associated with lower Q.
- Coles, Daniel, and Naveen (2008) account for heterogeneity and find that Q increases (decreases) in board size for complex (simple) firms.

The empirical setup: Endogeneity

- Does ‘good’ corporate governance cause higher valuations?
- Or are firms with higher market values more likely to choose better governance structures, e.g., because they have better investment opportunities and rely more heavily on external financing?
- Alternatively, firms may adopt good corporate governance to signal that firm insiders behave well, but governance is only a proxy for an omitted variable (for example, management quality) that simultaneously determines governance and performance.
- Similarly, firms with lower firm valuation may attempt to limit shareholders’ rights by adopting more G-Index provisions in order to insulate the firm from hostile takeovers.

The empirical setup: Endogeneity (II)

How can researchers mitigate such endogeneity concerns?

1) Econometric approaches:

- Firm fixed effects
- Instrumental variables regressions
- Heckman selection models
- Dynamic panel GMM models

2) Natural experiments:

- Example: Changes in regulation, e.g., Business Combination Laws, which protect firms from hostile takeovers and thereby increase the CEO's power.

CEO turnover and director reputation – Motivation

- Von Meyerinck, Romer, and Schmid (2024) analyze whether directors gain or lose reputation from forcing out a CEO.
- Two conflicting views:
 - 1) Firing a poorly performing CEO may be a sign of effective monitoring by the board.
 - This view represents the predominant position taken by the extant empirical finance literature (e.g., Farrell & Whidbee, 2000, JB; Faleye, 2007, JFE; Jenter & Kanaan, 2015, JF; Kempf et al., 2017, RFS).
 - Directors are expected to gain reputation

CEO turnover and director reputation – Motivation (II)

2) Firing a poorly performing CEO may cause disruption and/or indicate that the board did not replace the CEO before the negative performance became observable.

- This view is consistent with work in the management literature and theoretical work in financial economics (e.g., Adams and Ferreira, 2007, JF; Dow, 2013, JFE; Aghamolla and Hashimoto, 2021, JAR).

→ Directors are expected to lose reputation

⇒ Answering the question whether directors gain or lose reputation from forcing out a CEO, allows us to shed light on the corporate governance signal transmitted by a forced CEO turnover.

CEO turnover and director reputation – Empirical challenges

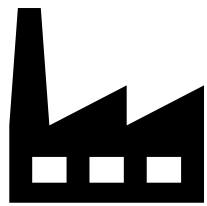
1. Turnover decisions are endogenous:

- We study directors with multiple directorships who are interlocked to a forced CEO turnover:
 - Interlocked firms, and the directors sitting on their boards, are largely unaffected by characteristics of the turnover firm, including factors that led to the forced CEO turnover.

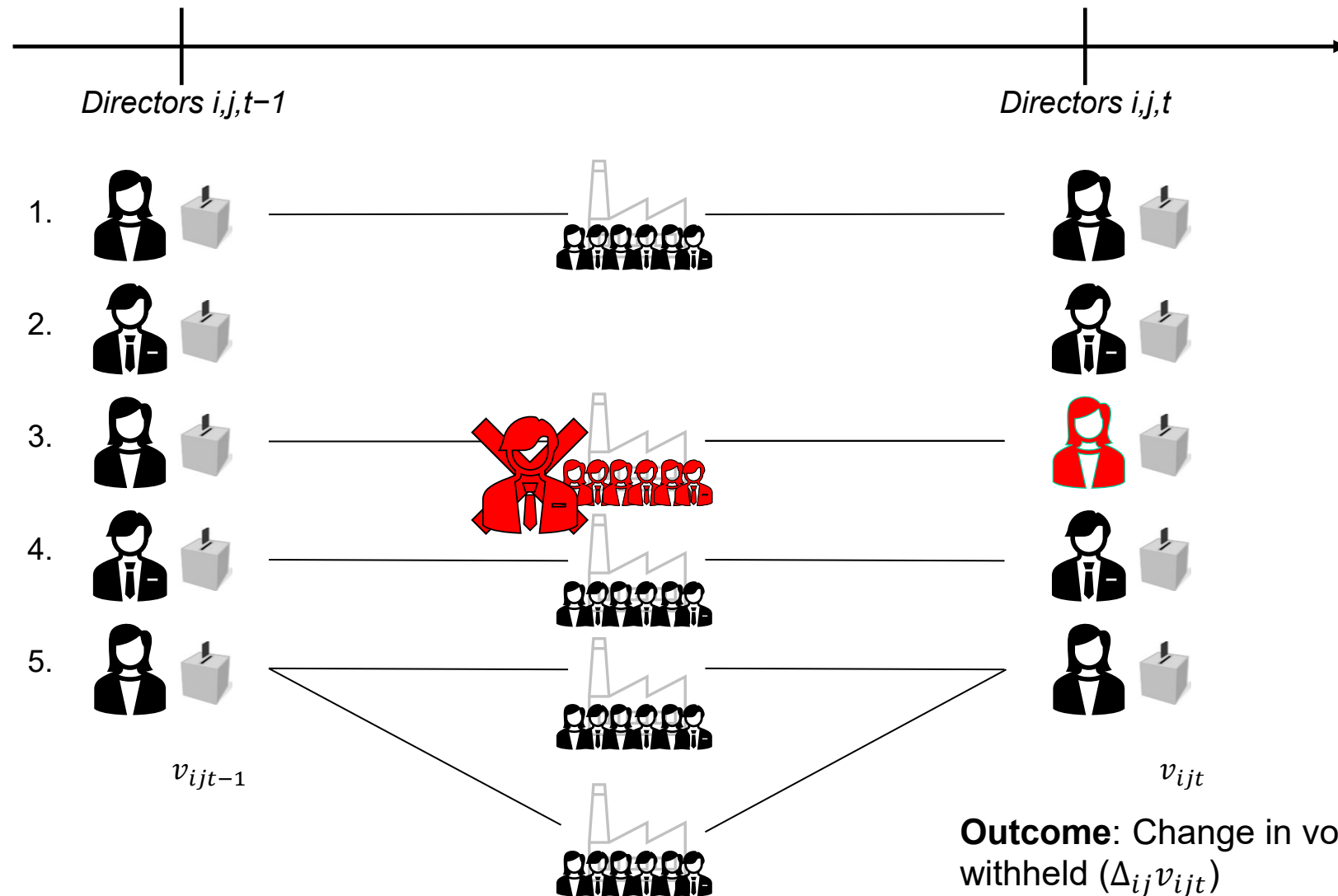
2. Widely used measures of director reputation are subject to endogenous selection:

- We use changes in withheld votes as a measure of director reputation:
 - Votes are not under directors' direct control.
 - Investors use withheld votes to evaluate directors' actions and directors respond to changes in withheld votes (e.g., Brochet & Srinivasan, 2014, JFE; Aggarwal et al., 2019, JFE; Erel et al., 2021, RFS).

CEO turnover and director reputation – Setting



Firm j



CEO turnover and director reputation – Empirical setting

- Difference-in-difference regressions at the director-firm-year level:

$$\Delta_{ij}v_{ijt} = \beta T_{ijt,t-1} + \delta \Delta_{ij}X_{it} + \theta \Delta_{ij}Z_{jt} + \alpha_{st} + \varepsilon_{ijt},$$

where i, j, s , and t index director, firm, industry, and years,

Δ is the first-difference operator,

v is the outcome variable (vote outcomes, board seats gained/lost)

T is the treatment indicator,

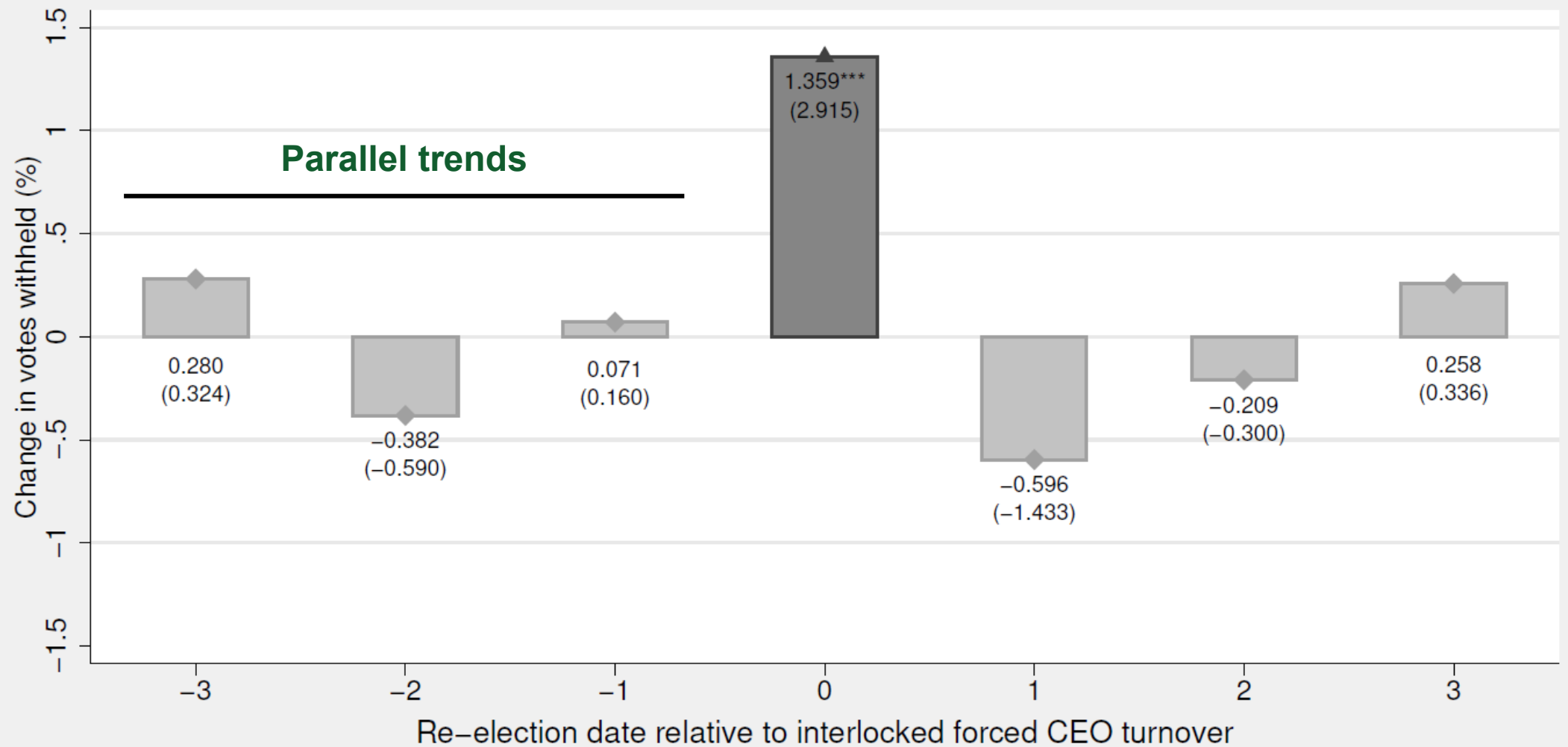
X and Z are time-varying director- and firm-level controls, and

α are fixed effects (baseline: industry x year)

CEO turnover and director reputation – Sample & data

- **BoardEx:** 1,773 CEO departures at S&P 1500 firms in the time period 2003-2017
- **Factiva:** Classify CEO turnovers as “forced” or “unforced” (Parrino, 1997, JFE; Peters & Wagner, 2014, JF; Jenter & Kanaan, 2015, JF)
- **ISS Voting Analytics:** Change in withheld votes from one shareholder meeting to the next
- **Sample:**
 - Treatment: 607 outside director re-elections interlocked to forced CEO turnovers
 - Control: 87,799 director re-elections

CEO turnover and director reputation – Univariate results



CEO turnover and director reputation – Multivariate results

- The effect of forced CEO turnover on interlocked vote shares:

Dependent variable:	% votes withheld		
	(1)	(2)	(3)
Forced interlocked (d)	1.204*** (3.071)	0.878** (2.553)	1.034** (2.407)
Control variables	Yes	Yes	Yes
Year $\times$ Industry FE	Yes	Yes	Yes
Firm FE	No	Yes	No
Director FE	No	No	Yes
Observations	88,406	88,406	88,406
Firms	3,269	3,269	3,269
Directors	18,693	18,693	18,693
Turnover-interlocked directorships	607	607	607
Adjusted R^2	0.333	0.347	0.354

Contextualizing the economic magnitude of results

Dependent variable:	% votes withheld			
	(1)	(2)	(3)	(4)
Restatement interlocked (d)	0.508 (1.617)			
Class action lawsuit interlocked (d)		0.854*** (4.515)		
Poison pill interlocked (d)			0.875** (2.077)	
Bankruptcy interlocked (d)				2.404** (2.102)
Control variables	Yes	Yes	Yes	Yes
Year $\times$ Industry FE	Yes	Yes	Yes	Yes
Observations	76,988	87,595	87,820	88,406
Firms	3,105	3,264	3,266	3,269
Directors	17,227	18,644	18,654	18,693
Events	330	217	139	8
Interlocked directorships	905	770	402	33
Adjusted R^2	0.405	0.333	0.328	0.333

CEO turnover and director reputation

What do these fixed effects do?

- Col. 1: Year x industry FE:
The turnover-interlocked directors are compared to other directors in the same industry and year.
- Col. 2: Year x industry FE plus firm FE:
Turnover-interlocked directors are compared to other non-turnover-interlocked directors at the same firm.
- Col. 3: Year x industry FE plus director FE:
Director are compared to themselves at times with and without an interlocked CEO turnover.

CEO turnover and director reputation – Cross-sectional results

Cross-sectional results show that these results are confined to:

- Forced CEO turnovers **without a full replacement announced**.
- Forced CEO turnovers that are **performance induced**.
- Forced CEO turnovers taking place at CEOs' **most productive tenure range** of 3-13 years (Brochet et al., 2021, TAR).
- Forced CEO turnovers in which the interlocked director has a **committee-based monitoring role** towards the CEO.
- Forced CEO turnovers where the turnover interlocked director is **affiliated with the CEO** (Coles et al., 2014, RFS).
- Directors who **did not leave the turnover firm** prior to the forced CEO turnover.

CEO turnover and director reputation – Omitted variables

- **Forced CEO turnovers and increases in withheld votes could result from poor performance of the turnover firms:**
 - **Propensity Score Matching (PSM) of turnover firms to non-turnover firms** (Peters and Wagner, 2014, JF)
 - Idea: Obtain balanced treatment and control samples of directors interlocked to firms that are similar in terms of the likelihood of firing the CEO but differ only in the effective turnover decision
 - Covariates: Company financial (valuation, stock returns, volatility,...), CEO characteristics, board characteristics, governance characteristics (compensation, G-Index, etc.).
- **Performance of the turnover firm does not constitute an omitted variable.**

CEO turnover and director reputation – Omitted variables (II)

- **Is there an unknown reason why, in two firms with equally bad performance, one chooses to fire its CEO and the other does not?**
 - If a management failure drives the bad performance, this may cause both CEO turnover and a negative updating about director ability.
 - If bad performance is due to bad luck, this may not convey any signal about directors' ability.
 - **Augment PSM analysis with a newspaper-based sentiment index:**
 - Mistakes of the leadership team trigger more negative newspaper coverage than bad luck.
 - This matched sample is balanced in terms of managers' involvement in negative events.
- **Results remain unchanged:** An unobservable variable related to management failures versus bad luck does not cause an OVB.

CEO turnover and director reputation – Alternative explanations

- **Alternative explanation: Director distraction** (e.g., Masulis & Zhang, 2019, JFE)
 - A forced CEO turnover demands significant time from involved directors and directors may neglect duties at interlocked firms.
- **Results are not driven by director distraction:**
 1. No increase in withheld votes following sudden CEO deaths at interlocked firms.
 2. No reduction in board meeting attendance for turnover-interlocked directors.

CEO turnover and director reputation – Labor market

- Do the negative reputational effects transmit into the director labor market?
- Outside directors involved in a forced CEO turnover on average do not lose board seats five years after the turnover.
- Initially lost directorships at the turnover firms are offset by newly acquired board seats in the subsequent four years.
- Lost board seats are replaced by new board seats at smaller firms.
- Smaller companies are associated with smaller compensation packages (e.g., Ryan & Wiggins, 2004), less power and prestige (e.g., Masulis & Mobbs, 2014), and reduced networking benefits (e.g., Yermack, 2004).

CEO turnover and director reputation – Publication

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CEO turnover and director reputation[☆]

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ABSTRACT

This paper analyzes the reputational effects of forced CEO turnovers on outside directors. We find that directors interlocked to a forced CEO turnover experience large and persistent increases in withheld votes at subsequent re-elections relative to non-turnover-interlocked directors. Directors are not penalized for an involvement in a turnover per se but for forced CEO turnovers that are related to governance failures by the board. Our results challenge the widespread view that forcing out a CEO can generally be understood as a sign of a well-functioning corporate governance.

Literature

Mandatory readings

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Additional readings

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